State-Space – Worked Problems

Practice: controllability and observability of given systems.

Ogata, *Modern Control Engineering*, Ch. 9 (end-of-chapter style).

Step through with **Ctrl+Enter**, or render a report with `publish`.

Contents

- Problem 1: Test Controllability and Observability
- Problem 2: Find the Value of a Parameter that Destroys Controllability
- Problem 3: Observability of a Sensor Placement Choice
- Problem 4: Controllability/Observability of a Transfer Function Realization
- Visualizing Problem 4: Pole-Zero Map
- Try it yourself

Problem 1: Test Controllability and Observability

For $\dot{x} = Ax + Bu,\ y = Cx$ with

$$A = [\,0\ \ 1\ \ 0;\ 0\ \ 0\ \ 1;\ -6\ \ -11\ \ -6\,],\quad B = [\,0;\,0;\,1\,],\quad C = [\,1\ \ 0\ \ 0\,]$$

determine complete state controllability and observability.

```
A1 = [0 1 0; 0 0 1; -6 -11 -6];
B1 = [0;0;1];
C1 = [1 0 0];

Co1 = ctrb(A1,B1);
Ob1 = obsv(A1,C1);
fprintf('Problem 1: rank(Co) = %d, rank(Ob) = %d (n=%d)\n', ...
    rank(Co1), rank(Ob1), size(A1,1))
fprintf('Controllable: %d, Observable: %d\n', ...
    rank(Co1)==size(A1,1), rank(Ob1)==size(A1,1))
```

```
Problem 1: rank(Co) = 3, rank(Ob) = 3 (n=3)
Controllable: 1, Observable: 1
```

Problem 2: Find the Value of a Parameter that Destroys Controllability

For

$$A = [\,-1\ \ 0;\ 0\ \ -2\,],\quad B = [\,1;\,\beta\,]$$

find all β for which the system is **not** controllable.

```
syms beta
A2 = [-1 0; 0 -2];
B2 = sym([1; beta]);
Co2 = [B2 A2*B2];
detCo2 = simplify(det(Co2));
fprintf('\nProblem 2: det(Co) = %s\n', char(detCo2))
beta_sol = solve(detCo2==0,beta);
fprintf('System is uncontrollable when beta = %s\n', char(beta_sol))
```

```
Problem 2: det(Co) = -beta
System is uncontrollable when beta = 0
```

Problem 3: Observability of a Sensor Placement Choice

For the same A as Problem 2 with two candidate single-output matrices $C_a = [1\ 0]$ and $C_b = [1\ 1]$, determine which sensor placement(s) preserve observability.

```
A3 = [-1 0; 0 -2];
Ca = [1 0];
Cb = [1 1];
fprintf('\nProblem 3: rank(obsv(A,Ca)) = %d\n', rank(obsv(A3,Ca)))
fprintf('Problem 3: rank(obsv(A,Cb)) = %d\n', rank(obsv(A3,Cb)))
fprintf('Ca alone misses mode at s=-2 (decoupled, unmeasured); Cb observes both.\n')
```

```
Problem 3: rank(obsv(A,Ca)) = 1
Problem 3: rank(obsv(A,Cb)) = 2
Ca alone misses mode at s=-2 (decoupled, unmeasured); Cb observes both.
```

Problem 4: Controllability/Observability of a Transfer Function Realization

For $G(s) = \frac{s+2}{(s+1)(s+3)(s+4)}$, realize in controllable canonical form and verify controllability is automatic (by construction) while checking observability (which can fail if there is pole-zero cancellation).

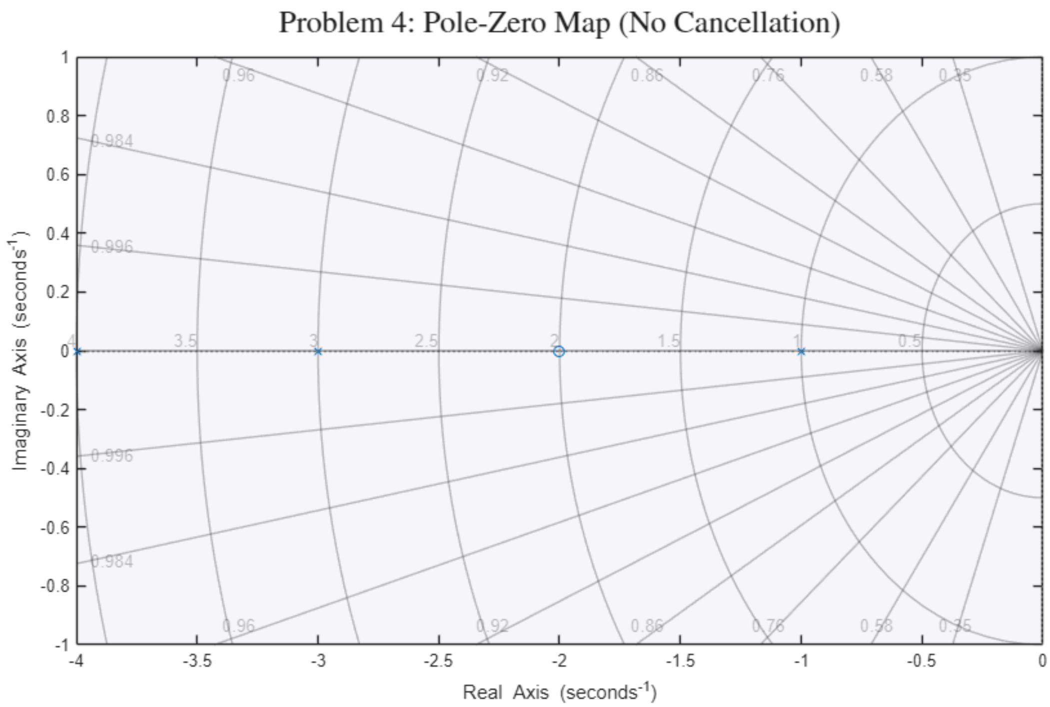
```
G4 = tf([1 2],conv(conv([1 1],[1 3]),[1 4]));
sys4 = ss(G4);
A4 = sys4.A; B4 = sys4.B; C4 = sys4.C;
fprintf('\nProblem 4: rank(Co) = %d, rank(Ob) = %d (n=%d)\n', ...
    rank(ctrb(A4,B4)), rank(obsv(A4,C4)), size(A4,1))
fprintf('No pole-zero cancellation present, so both hold (n=%d, full rank).\n', size(A4,1))
```

```
Problem 4: rank(Co) = 3, rank(Ob) = 3 (n=3)
No pole-zero cancellation present, so both hold (n=3, full rank).
```

Visualizing Problem 4: Pole-Zero Map

The realization of $G(s) = \frac{s+2}{(s+1)(s+3)(s+4)}$ has its zero at -2 distinct from every pole, so no pole-zero cancellation occurs and the system is a minimal realization – both controllable and observable.

```
figure
pzmap(sys4)
grid on
title('Problem 4: Pole-Zero Map (No Cancellation)','Interpreter','latex','FontSize',18)
```



Try it yourself

- In Problem 2, change `A` and re-solve `detCo2==0` to find that system's own uncontrollable `beta`.
- In Problem 4, force a pole-zero cancellation (put the zero at -1) and watch observability drop below full rank.